



Métodos y programas informáticos en biomedicina

Apoya el acceso abierto

Guía para autores

Acerca de la revista

- [Objetivos y alcance](#)
- [Anuncios de la revista](#)
- [Revisión por pares](#)
- [Acceso abierto](#)

Ética y políticas

- [Ética en la publicación](#)
- [Declaración de presentación](#)
- [Paternidad literaria](#)
- [Cambios en la autoría](#)
- [Declaración de intereses contrapuestos](#)
- [Fuentes de financiación](#)
- [Declaración sobre el uso de IA generativa](#)
- [Preimpresiones](#)
- [Uso de lenguaje inclusivo](#)
- [Informes de análisis basados en el sexo y el género](#)
- [Reclamaciones jurisdiccionales](#)
- [Estudios en humanos](#)
- [Ensayos clínicos](#)
- [Publicación retrasada](#)
- [Retirada, corrección o retractación](#)

Redacción y formato

- [Formato de archivo](#)
- [Página de título](#)
- [Abstracto](#)
- [Palabras clave](#)
- [Reflejos](#)
- [fórmulas matemáticas](#)
- [Tablas](#)
- [Figuras, imágenes y obras de arte](#)
- [Inteligencia artificial generativa y figuras, imágenes y otras obras de arte.](#)
- [Material complementario](#)
- [Video](#)
- [Datos de investigación](#)

- [Declaración de datos](#)
- [Vinculación de datos](#)
- [Presentación conjunta de datos, métodos o protocolos relacionados.](#)
- [Información específica de la revista: Redacción y formato](#)
- [Estructura del artículo](#)
- [Referencias](#)

Envío de su manuscrito

- [Lista de verificación para la presentación](#)
- [Enviar en línea](#)

Tras recibir una decisión final

- [Servicio de transferencia de artículos](#)
- [Acuerdo de publicación](#)
- [Opciones de licencia](#)
- [Acceso abierto](#)
- [Permiso para obras protegidas por derechos de autor](#)
- [Corrección de pruebas](#)
- [Compartir enlace](#)
- [Compartir de forma responsable](#)

Recursos para autores

- [Academia de Investigadores de Elsevier](#)
- [Servicios de idiomas y edición](#)

Obtener ayuda y apoyo

- [Soporte al autor](#)

Acerca de la revista

Objetivos y alcance

Fomentar el desarrollo de **métodos informáticos** formales y su aplicación en la investigación biomédica y la práctica médica, mediante la ilustración de principios fundamentales en la investigación **en bioinformática** ; estimular la investigación básica en el diseño de software de aplicación; informar sobre el estado de la investigación de proyectos **de procesamiento de información biomédica** ; informar sobre nuevas metodologías informáticas aplicadas en áreas biomédicas; la eventual distribución de software demostrable para evitar la duplicación de esfuerzos; proporcionar un foro para la discusión y mejora del software existente; optimizar el contacto entre organizaciones nacionales y grupos de usuarios regionales promoviendo un intercambio internacional de información sobre métodos formales, estándares y software en biomedicina.

Métodos y programas informáticos en biomedicina abarca la metodología informática y **los sistemas de software** derivados de la informática para su aplicación en todos los aspectos de la investigación biomédica y la práctica médica. Está diseñado para servir a: bioquímicos; biólogos; genetistas; inmunólogos; neurocientíficos; farmacólogos; toxicólogos; médicos; epidemiólogos; psiquiatras; psicólogos; cardiólogos; químicos; (radio)físicos; científicos informáticos; programadores y analistas de sistemas; ingenieros biomédicos, clínicos, eléctricos y de otras disciplinas; profesores de informática médica y usuarios de software educativo.

Computer Methods and Programs in Biomedicine es el título complementario de la revista de acceso abierto [Computer Methods and Programs in Biomedicine Update](#).

Anuncios de la revista

Ya no se aceptan manuscritos con varios autores de correspondencia. Si su manuscrito aceptado tiene más de un autor de correspondencia, el artículo publicado incluirá únicamente al autor registrado en Editorial Manager o, en su defecto, al autor de correspondencia principal del manuscrito.

Declaración ética

La revista exige que los autores confirmen al momento de la presentación si su investigación requiere una declaración ética. Dicha declaración confirmará que todos los procedimientos se realizaron de conformidad con las leyes y directrices institucionales pertinentes y que fueron aprobados por el comité o comités institucionales correspondientes. Esta declaración debe incluir la fecha y el número de referencia de la aprobación ética obtenida. Los autores también deben incluir en la declaración ética que se obtuvo el consentimiento informado para la experimentación con sujetos humanos. Los derechos de privacidad de los sujetos humanos deben respetarse siempre. En caso de que su investigación no requiera una declaración ética, deben explicar el motivo.

Revisión por pares

Esta revista sigue un proceso de revisión único y anónimo. Su manuscrito será evaluado inicialmente por nuestros editores para determinar su idoneidad para su publicación. Si se considera adecuado, se enviará a un mínimo de dos revisores para una evaluación independiente y experta de su calidad científica. La decisión final sobre la aceptación o el rechazo del artículo la tomarán nuestros editores.

Lea más sobre [la revisión por pares](#).

Nuestros editores no participan en la toma de decisiones sobre los artículos que:

- Se han escrito ellos mismos.
- Han sido escritos por miembros de la familia o compañeros de trabajo.
- relacionarse con productos o servicios en los que tienen algún interés.

Todos los trabajos presentados estarán sujetos a los procedimientos habituales de la revista y la revisión por pares se realizará de forma independiente del editor y su grupo de investigación. Lea más sobre [las funciones del editor](#).

Los autores pueden presentar una apelación formal contra la decisión editorial, siempre que cumpla con todos los requisitos y siga el procedimiento descrito en [la Política de Apelaciones](#) de Elsevier. Solo se considerará una apelación por manuscrito y la decisión será definitiva.

Ediciones especiales y colecciones de artículos

El proceso de revisión por pares para números especiales y colecciones de artículos sigue el mismo procedimiento descrito anteriormente para las presentaciones regulares, con la excepción de que un editor invitado puede enviar los artículos a los revisores y recomendar una decisión al editor de la revista. El editor de la revista supervisa el proceso de revisión por pares de todos los números especiales y colecciones de artículos para garantizar que se respeten los altos estándares de ética editorial y capacidad de respuesta, y es responsable de la decisión final sobre la aceptación o el rechazo de los artículos.

Acceso abierto

Para obtener información sobre las opciones de acceso abierto para esta revista, le recomendamos que consulte nuestra [página de información sobre acceso abierto](#).

Ética y políticas

Ética en la publicación

Los autores deben seguir las directrices éticas establecidas en [la Política de Ética Editorial](#) de Elsevier .

Declaración de presentación

Cuando los autores envían un artículo a una revista de Elsevier, se sobreentiende que:

- El trabajo descrito no ha sido publicado previamente, salvo en forma de preimpresión, resumen, conferencia publicada, tesis académica o informe registrado. Consulte nuestra política sobre [publicaciones múltiples, redundantes o simultáneas](#) .
- El artículo no está siendo considerado para su publicación en ningún otro medio.
- La publicación del artículo cuenta con la aprobación de todos los autores y, tácita o explícitamente, de las autoridades responsables del lugar donde se realizó el trabajo.
- En caso de ser aceptado, el artículo no podrá ser publicado en ningún otro medio, en la misma forma, ni en inglés ni en ningún otro idioma, incluyendo formatos electrónicos, sin el consentimiento por escrito del titular de los derechos de autor.

Para verificar el cumplimiento de nuestras políticas de publicación de la revista, podemos revisar su manuscrito con nuestras herramientas de selección.

Paternidad literaria

Todos los autores deben haber realizado contribuciones sustanciales a todos los siguientes aspectos:

1. La concepción y el diseño del estudio, o la adquisición de datos, o el análisis e interpretación de los datos.
2. Redactar el artículo o revisarlo críticamente para garantizar un contenido intelectual importante.
3. Aprobación final de la versión que se va a presentar.

Todos los autores son responsables de todos los aspectos del trabajo, para garantizar que las cuestiones relacionadas con la exactitud o la integridad de cualquier parte del mismo sean debidamente investigadas y resueltas.

Los autores deberán designar a un único autor de correspondencia para comunicarse con la revista durante el proceso editorial. Esta persona enviará el manuscrito y toda la documentación requerida a la revista, y será el principal punto de contacto entre la revista y los coautores. Únicamente la afiliación del autor de correspondencia se utilizará para determinar la elegibilidad para un contrato de publicación y los posibles descuentos relacionados. Las afiliaciones de los demás coautores no son relevantes para la elegibilidad.

Cambios en la autoría

Por lo general, los editores de esta revista no considerarán cambios en la autoría una vez que se haya enviado un manuscrito. Es importante que los autores revisen cuidadosamente la lista de autores y el orden de los mismos, y que proporcionen una lista definitiva al momento del envío.

La política de esta revista con respecto a los cambios de autoría es la siguiente:

- Todos los autores deben figurar en el manuscrito y sus datos deben ingresarse en el sistema de envío. Solo se podrán realizar cambios antes de la aceptación y únicamente con la aprobación del editor de la revista. Esto incluye la adición, eliminación o reordenación de los nombres de los autores.
- Las solicitudes de cambio de autoría deben ser realizadas por el autor correspondiente mediante el [formulario de solicitud de cambio de autoría](#). El autor correspondiente debe proporcionar al editor de la revista el motivo de la solicitud, junto con una confirmación por escrito de todos los autores, incluidos aquellos que se añadan o eliminen, de que están de acuerdo con los cambios. Las solicitudes que no cumplan con las instrucciones del formulario no serán consideradas.
- Esta revista no permite cambios en la autoría después de la aceptación del artículo. Esto incluye adiciones, eliminaciones o cambios en el orden de los nombres de los autores, incluidos los cambios en el autor de correspondencia.
- El proceso de revisión puede suspenderse mientras se considera una solicitud de cambio de autoría.
- Las solicitudes de cambio realizadas por los autores sin el [formulario de solicitud de cambio de autoría](#) o sin la aprobación del editor pueden dar lugar al rechazo del manuscrito o a su retractación si el artículo ya ha sido publicado.

Declaración de intereses contrapuestos

Todos los autores deben revelar cualquier relación financiera o personal con otras personas u organizaciones que pudiera influir o sesgar indebidamente su trabajo. Algunos ejemplos de posibles conflictos de intereses son:

- Empleo
- Consultorías
- Propiedad de acciones
- Honorarios
- Testimonio pericial remunerado
- Solicitudes o registros de patentes
- Subvenciones o cualquier otro tipo de financiación
- Afiliación a la revista como editor o miembro del consejo asesor.

La [herramienta de declaraciones](#) siempre debe completarse.

Los autores que tengan una afiliación a una revista que declarar deben ingresar el siguiente texto en "Otras actividades" dentro de la [herramienta de declaraciones](#) y deben informar a la revista y al editor antes de completar el proceso de envío:

Dado su rol como [CARGO DEL AUTOR EN LA REVISTA], [NOMBRE DEL AUTOR] no participó en la revisión por pares de este artículo y no tiene acceso a información sobre dicha revisión. La responsabilidad total del proceso editorial de este artículo fue delegada a otro editor de la revista.

Las declaraciones de divulgación editorial se incluirán como nota al pie y/o en la sección de declaración de conflictos de intereses del artículo.

Los autores que no tengan conflictos de intereses que declarar deben seleccionar la opción "No tengo nada que declarar".

El documento de Word resultante, que contiene su declaración, debe cargarse en el paso "Adjuntar/Subir archivos" del proceso de envío. Es importante que el documento de Word se guarde en formato .doc/.docx. No se requieren firmas de los autores.

Fuentes de financiación

Los autores deben revelar cualquier fuente de financiación que haya proporcionado apoyo económico para la realización de la investigación y/o la preparación del artículo. El papel de los patrocinadores, si los hubiere, debe declararse en relación con el diseño del estudio, la recopilación, el análisis y la interpretación de los datos, la redacción del informe y la decisión de enviar el artículo para su publicación. Si las fuentes de financiación no tuvieron ninguna participación, esto debe indicarse en el manuscrito.

Enumere las fuentes de financiación de esta forma estándar para facilitar el cumplimiento de los requisitos del financiador:

Financiación: Este trabajo fue financiado por los Institutos Nacionales de Salud [números de subvención xxxx, yyyy]; la Fundación Bill y Melinda Gates, Seattle, WA [número de subvención zzzz]; y los Institutos de Paz de los Estados Unidos [número de subvención aaaa].

No es necesario incluir descripciones detalladas del programa ni del tipo de becas, subvenciones o premios. Si la financiación proviene de una subvención global u otros recursos disponibles para una universidad, colegio u otra institución de investigación, indique el nombre de la institución u organización que la proporcionó.

Si no se ha proporcionado financiación para la investigación, se recomienda incluir la siguiente frase:

Esta investigación no recibió ninguna subvención específica de agencias de financiación de los sectores público, comercial o sin ánimo de lucro.

Declaración sobre el uso de IA generativa

Los autores deben declarar el uso de herramientas de IA generativa en el proceso de preparación del manuscrito al momento de enviar el artículo.

La IA generativa y las tecnologías asistidas por IA ("herramientas de IA") se han adoptado rápidamente en la investigación y la publicación académica, ofreciendo beneficios significativos al tiempo que requieren estándares claros para un uso responsable y transparente.

Los autores que preparen un manuscrito para esta revista pueden utilizar herramientas de IA como apoyo. Sin embargo, estas herramientas nunca deben sustituir el pensamiento crítico, la experiencia ni la evaluación humana. Las herramientas de IA solo pueden aplicarse bajo supervisión y control humano.

Por favor, lea la política detallada de Elsevier para autores sobre el uso de herramientas de IA, tal como se indica en [las políticas de IA generativa de Elsevier para revistas](#).

En última instancia, los autores son responsables del contenido de su obra. Esto incluye la responsabilidad por:

- Revisar y verificar cuidadosamente la exactitud, la exhaustividad y la imparcialidad de todos los resultados generados por la IA (incluida la comprobación de las fuentes, ya que las referencias generadas por la IA pueden ser incorrectas o falsas).
- Editar y adaptar minuciosamente todo el material para garantizar que el manuscrito represente la contribución auténtica y original del autor y refleje su propio análisis, interpretación, perspectivas e ideas.
- Garantizar la exactitud y originalidad de las imágenes generadas por IA y proporcionar la atribución correspondiente cuando proceda.
- Ensuring the use of any tools or sources, AI-based or otherwise, is made clear and transparent to readers. If AI tools have been used, we require a disclosure statement upon submission; please see example below.
- Ensuring the manuscript is developed in a way that safeguards data privacy, intellectual property and other rights, by checking the terms and conditions of any AI tool that is used.

Finally, authors must not list or cite AI tools as an author or co-author on the manuscript since authorship implies responsibilities and tasks that can only be attributed to, and performed by, humans.

The use of AI tools in the manuscript preparation process must be declared by adding a statement at the end of the manuscript when the paper is first submitted. The statement will appear in the published work and should be placed in a new section before the references list. An example:

Title of new section: *Declaration of generative AI and AI-assisted technologies in the manuscript preparation process.*

Statement: *During the preparation of this work the author(s) used [NAME OF TOOL / SERVICE] in order to [REASON]. After using this tool/service, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the published article.*

The declaration does not apply to the use of basic tools, such as tools used to check grammar, spelling and references. In addition, authors are not required to disclose the use of AI tools or features within specialist disability-related assistive technology, provided these are used solely for accessibility purposes.

If you have nothing to disclose, you do not need to add a statement.

Please note: to protect authors' rights and the confidentiality of their research, this journal does not currently allow academic editors and reviewers to upload an unpublished manuscript into a generative AI tool, as is stated in Elsevier's [generative AI policies for journals](#). However, AI tools may be used in a supportive capacity by editors and reviewers to improve structure and language of the review report or editorial decision letter or to conduct a background literature search. Elsevier is actively evaluating compliant AI tools and may revise this policy in the future.

Preprints

Preprint sharing

Authors may share preprints in line with Elsevier's [article sharing policy](#). Sharing preprints, such as on a preprint server, will not count as prior publication.

We advise you to read our policy on [multiple, redundant or concurrent publication](#).

In support of [Open Science](#), this journal offers its authors a free preprint posting service on [SSRN](#), Elsevier's preprint and early research repository.

During submission to this journal, you can choose to post your manuscript on SSRN, and it will be made publicly available immediately.

As a preprint on SSRN, your manuscript will benefit from:

- Early registration with a preprint DOI (Digital Object Identifier)
- A link from the preprint to the version of record if published in this journal
- Preprint posting, sharing, and download availability that facilitates collaboration and early citations

Your decision to post or not post your preprint will have no effect on the editorial process or publication outcome with the journal. For additional information, please consult the [SSRN Terms of Use](#) and [FAQs](#).

It is expected that the corresponding author will seek approval from all co-authors before agreeing to post the manuscript publicly, prior to peer review, on SSRN.

Use of inclusive language

Inclusive language acknowledges diversity, conveys respect to all people, is sensitive to differences, and promotes equal opportunities. Authors should ensure their work uses inclusive language throughout and contains nothing which might imply one individual is superior to another on the grounds of:

- age
- gender
- race
- ethnicity
- culture
- sexual orientation
- disability or health condition

We recommend avoiding the use of descriptors about personal attributes unless they are relevant and valid. Write for gender neutrality with the use of plural nouns ("clinicians, patients/clients") as default. Wherever possible, avoid using "he, she," or "he/she."

No assumptions should be made about the beliefs of readers and writing should be free from bias, stereotypes, slang, reference to dominant culture and/or cultural assumptions.

These guidelines are meant as a point of reference to help you identify appropriate language but are by no means exhaustive or definitive.

Reporting sex- and gender-based analyses

There is no single, universally agreed-upon set of guidelines for defining sex and gender. We offer the following guidance:

- Sex and gender-based analyses (SGBA) should be integrated into research design when research involves or pertains to humans, animals or eukaryotic cells. This should be done in accordance with any requirements set by funders or sponsors and best practices within a field.
- Sex and/or gender dimensions of the research should be addressed within the article or declared as a limitation to the generalizability of the research.
- Definitions of sex and/or gender applied should be explicitly stated to enhance the precision, rigor and reproducibility of the research and to avoid ambiguity or conflation of terms and the constructs to which they refer.

We advise you to read the [Sex and Gender Equity in Research \(SAGER\) guidelines](#) and the [SAGER checklist](#) (PDF) on the EASE website, which offer systematic approaches to the use of sex and gender information in study design, data analysis, outcome reporting and research interpretation.

For further information we suggest reading the rationale behind and recommended [use of the SAGER guidelines](#).

Definitions of sex and/or gender

We ask authors to define how sex and gender have been used in their research and publication. Some guidance:

- Sex generally refers to a set of biological attributes that are associated with physical and physiological features such as chromosomal genotype, hormonal levels, internal and external anatomy. A binary sex categorization (male/female) is usually designated at birth ("sex assigned at birth") and is in most cases based solely on the visible external anatomy of a newborn. In reality, sex categorizations include people who are intersex/have differences of sex development (DSD).
- Gender generally refers to socially constructed roles, behaviors and identities of women, men and gender-diverse people that occur in a historical and cultural context and may vary across societies and over time. Gender influences how people view themselves and each other, how they behave and interact and how power is distributed in society.

Depending on the focus of a paper, sex and/or gender may or may not be relevant to the content of the paper. We recognize that beliefs, attitudes, and laws relating to sex and gender may vary. These articles do not attempt to dictate author beliefs but rather require that, where relevant to an author's research or paper, the author must provide clear explanations of how the paper and research define and use sex and gender.

Jurisdictional claims

Elsevier respects the decisions taken by its authors as to how they choose to designate territories and identify their affiliations in their published content. Elsevier's policy is to take a neutral position with respect to territorial disputes or jurisdictional claims, including, but not limited to, maps and institutional affiliations. For journals that Elsevier publishes on behalf of a third party owner, the owner may set its own policy on these issues.

- Maps: Readers should be able to locate any study areas shown within maps using common mapping platforms. Maps should only show the area actually studied and authors should not include a location map which displays a larger area than the bounding box of the study area. Authors should add a note clearly stating that "*map lines delineate study areas and do not necessarily depict accepted national boundaries*". During the review process, Elsevier's editors may request authors to change maps if these guidelines are not followed.
- Institutional affiliations: Authors should use either the full, standard title of their institution or the standard abbreviation of the institutional name so that the institutional name can be independently verified for research integrity purposes.

Studies in humans

Authors must follow [ethical guidelines](#) for research involving humans, their data, and/or their biological materials (e.g. organs, tissues). Authors should ensure that they adhere to the following:

- All studies must be performed in compliance with relevant laws, regulations, and institutional guidelines and that the appropriate institutional committee(s) has/have approved them. For instance, medical research involving humans should be carried out in accordance with the World Medical Association [Declaration of Helsinki: Ethical principles for medical research involving human participants](#) and should follow the International Committee of Medical Journal Editors (ICMJE) [recommendations for the conduct, reporting, editing and publication of scholarly work in medical journals](#) and aim to be representative of human populations in terms of sex, age and ethnicity. Sex and gender terms should be used correctly, as outlined in the "Reporting Sex- and Gender-based analyses" section of this journal's Guide for Authors.
- Informed consent must be obtained for research with human participants, including their data or biological materials. Consent must be given without coercion, and the privacy rights of human participants must always be observed. If your study contains human participants, please read Elsevier's full [human research policy](#) for further information.
 - Authors should be aware that written (or electronic) consent is required for certain studies under national laws. In studies where only verbal informed consent was obtained, authors should include justification for the absence of written consent, the name of the approving ethics committee, and a statement on how verbal consent was recorded.
 - Consent for publication should also be obtained.
- Authors must obtain and retain explicit written consent for any identifiable participant materials (e.g. case details, videos, recordings, images, photographs, and illustrations). All participant information must be de-identified: do not include names or personal identifiers (e.g., initials, ID numbers, DOB)
- Images should be used only when scientifically necessary and with explicit consent, and nonessential identifying details must be removed. If identifiers are altered to ensure anonymity, authors must confirm that scientific meaning is not affected. Fully anonymized images (e.g., x-rays, scans) that contain no identifying marks and are not

accompanied by identifying text do not require formal consent, though informing the patient is good practice. Case reports always require consent. Simply obscuring identity (e.g., eye bars or blurring) is not sufficient without consent.

- Written consents must be retained by the authors. They should not be provided to this journal unless this is specifically requested in exceptional circumstances, for example, when a legal issue arises. Only then should you provide copies of the consents, or evidence that all relevant consents were obtained.
- Appropriate consents and safeguards should be in place for research with vulnerable populations, as described in Elsevier's [human research policy](#). In many countries, additional ethical approval may be needed for research with those who lack the capacity to consent (e.g. children or those with mental health or neurological conditions including dementia).
- For the full policy, see "[Consent requirements for case reports, personal information, and images](#)."

All studies involving human participants require a statement that details any ethical approval(s), guidelines followed, confirmation of informed consent, and any other necessary information in line with Elsevier's [human research policy](#). The statement should contain the date and reference number of any ethical approval(s) obtained

If ethical approval is not required by national laws or where a study has been granted an exemption from an ethics committee, the statement must explain why the study is exempt and confirm that the appropriate protocols for protecting the rights and privacy of all participants were utilized during the execution of the research (e.g. no coercion to participate, full disclosure of study requirements and risks, consent of participants (or their Legal Representatives), no release of participant data without their knowledge, etc.).

Human biological material

For all studies that use human organs or tissues, sufficient evidence must be provided that these were procured in line with [WHO Guiding Principles on Human Cell, Tissue and Organ Transplantation](#). The source of the organs or tissues used in clinical research must be transparent and traceable and must not be sourced from executed prisoners or prisoners of conscience, consistent with recommendations by [Global Rights Compliance on Mitigating Human Rights Risks in Transplantation Medicine](#).

For clinical studies, a statement of informed consent having been obtained from a patient or their legal representative, paired with ethical approval for the study from a suitable institution, as required by the policies of the journal, may be considered sufficient evidence, but the journal reserves the right to request additional evidence in cases where it feels this is not sufficient.

If the manuscript describes organ transplantation authors must additionally declare within the manuscript that:

- Autonomous consent, free from coercion, was obtained from the donor(s) or their next of kin.
- Organs and/or tissues were not sourced from executed prisoners or prisoners of conscience.

Clinical Trials

This journal follows the [International Committee of Medical Journal Editors \(ICMJE\) clinical trials registration guidelines](#).

Registration

Clinical trials must be registered in a public trials registry. Purely observational studies, in which the assignment of the medical intervention is not at the discretion of the investigator, do not require registration.

Some important points:

- Trials must be registered at or before the onset of patient enrolment.
- The clinical trial registration number must be included in the manuscript, preferably at the end of the article abstract.
- A clinical trial is defined as any research study that prospectively assigns human participants, or groups of humans, to one or more health-related interventions to evaluate the effects of health outcomes.
- Health-related interventions include any intervention used to modify a biomedical or health-related outcome such as drugs, surgical procedures, devices, behavioral treatments, dietary interventions, and process-of-care changes.
- Health outcomes include any biomedical or health-related measures obtained in patients or participants, including pharmacokinetic measures and adverse events.

Results

Authors are required to disclose all posting in registries of results of the same or closely related work. Editors will not consider results to be a prior publication if they have already been posted in the same clinical trials registry in which primary registration resides, as long as the results are presented in the form of a brief structured abstract (fewer than 500 words) or table.

Disclosing results in other circumstances, such as in an investors' meeting, for example, is discouraged and may jeopardize consideration of your manuscript by this journal.

Reporting

Authors are encouraged to follow CONSORT guidelines when presenting randomized controlled trials, and provide the CONSORT checklist at manuscript submission, with an accompanying flow diagram illustrating the progress of patients through the trial - including recruitment, enrolment, randomization, withdrawal, completion and a description of the randomization procedure.

- Read the [CONSORT guidelines](#).
- Follow the [CONSORT checklist](#).

Delayed publication

If you need to delay the publication of your article for any reason, please contact the editorial office of the journal or our [Journal Article Publishing Support Center](#) at the earliest possible opportunity. While we will try to ensure that your article is not published before a certain date if you make your request on a timely basis, we cannot guarantee this. Please also note that once the agreed period of delay has expired the article may be published at any time thereafter.

Withdrawal, correction or retraction

For questions about article withdrawal, correction, or retraction, please consult Elsevier's [Article Correction, Retraction and Removal Policy](#).

Writing and formatting

File format

We ask you to provide editable source files for your entire submission (including figures, tables and text graphics). Some guidelines:

- Save files in an editable format, using the extension .doc/.docx for Word files and .tex for LaTeX files. A PDF is not an acceptable source file.
- Format Word files in a single-column layout. Double-column formatting is only permitted for LaTeX submissions.
- Remove any strikethrough and underlined text from your manuscript, unless it has scientific significance related to your article.
- Use spell-check and grammar-check functions to avoid errors.

We advise you to read our [Step-by-step guide to publishing with Elsevier](#).

Title page

You are required to include the following details in the title page information:

- Article title. Article titles should be concise and informative. Please avoid abbreviations and formulae, where possible, unless they are established and widely understood, e.g. DNA.
- Author names. Provide the given name(s) and family name(s) of each author. The order of authors should match the order in the submission system. Carefully check that all names are accurately spelled. If needed, you can add your name between parentheses in your own script after the English transliteration.
- Affiliations. Add affiliation addresses, referring to where the work was carried out, below the author names. Indicate affiliations using a lower-case superscript letter immediately after the author's name and in front of the corresponding address. Ensure that you provide the full postal address of each affiliation, including the country name and, if available, the email address of each author.
- Corresponding author. Clearly indicate who will handle correspondence for your article at all stages of the refereeing and publication process and also post-publication. This responsibility includes answering any future queries about your results, data, methodology and materials. It is important that the email address and contact details of your corresponding author are kept up to date during the submission and publication process.
- Present/permanent address. If an author has moved since the work described in your article was carried out, or the author was visiting during that time, a "present address" (or "permanent address") can be indicated by a footnote to the author's name. The address where the author carried out the work must be retained as their main affiliation address. Use superscript Arabic numerals for such footnotes.

Abstract

You are required to provide a concise and factual abstract which does not exceed 250 words. The abstract should briefly state the purpose of your research, principal results and major conclusions. Some guidelines:

- Abstracts must be able to stand alone as abstracts are often presented separately from the article.
- Avoid references. If any are essential to include, ensure that you cite the author(s) and year(s).
- Avoid non-standard or uncommon abbreviations. If any are essential to include, ensure they are defined within your abstract at first mention.

Keywords

You are required to provide 3 to 6 keywords for indexing purposes. Keywords should be written in English. Please try to avoid keywords consisting of multiple words (using "and" or "of").

We recommend that you only use abbreviations in keywords if they are firmly established in the field.

Highlights

You are encouraged to provide article highlights at submission.

Highlights are a short collection of bullet points that should capture the novel results of your research as well as any new methods used during your study. Highlights will help increase the discoverability of your article via search engines. Some guidelines:

- Submit highlights as a separate editable file in the online submission system with the word "highlights" included in the file name.
- Highlights should consist of 3 to 5 bullet points, each a maximum of 85 characters, including spaces.

We encourage you to view example [article highlights](#) and read about the benefits of their inclusion.

Math formulae

- Submit math equations as editable text, not as images.
- Present simple formulae in line with normal text, where possible.
- Use the solidus (/) instead of a horizontal line for small fractional terms such as X/Y.
- Present variables in italics.
- Denote powers of e by exp.
- Display equations separately from your text, numbering them consecutively in the order they are referred to within your text.

Tables

Tables must be submitted as editable text, not as images. Some guidelines:

- Place tables next to the relevant text or on a separate page(s) at the end of your article.
- Cite all tables in the manuscript text.
- Number tables consecutively according to their appearance in the text.
- Please provide captions along with the tables.
- Place any table notes below the table body.
- Avoid vertical rules and shading within table cells.

We recommend that you use tables sparingly, ensuring that any data presented in tables is not duplicating results described elsewhere in the article.

Figures, images and artwork

Figures, images, diagrams and other graphical media (artwork) must be supplied as separate files along with the manuscript, in line with these [artwork and media instructions](#). Some excerpts:

When submitting artwork:

- Cite all images in the manuscript text.
- Number images according to the order they appear within your article.
- Submit each image as a separate file using a logical naming convention for your files (for example, Figure_1, Figure_2 etc).
- Text graphics may be embedded in the text at the appropriate position. If you are working with LaTeX, text graphics may also be embedded in the file.

Captions

All artwork must have a caption. A caption should consist of a brief title (not displayed on the figure itself) and a description of the image. Keep the amount of text in any image to a minimum, though any symbol or abbreviation should be explained.

Read how to add captions to your submission [here](#).

Formats

When your artwork is finalized, "save as" or convert your electronic artwork to the formats listed below taking into account the given resolution requirements for line drawings, halftones, and line/halftone combinations:

- Vector drawings: Save as EPS or PDF files embedding the font or saving the text as "graphics."
- Color or grayscale photographs (halftones): Save as TIFF, JPG or PNG files using a minimum of 300 dpi (for single column width: min. 1063 pixels, full page width: 2244 pixels).

- Bitmapped line drawings: Save as TIFF, JPG or PNG files, using a minimum of 1000 dpi (for single column width: min. 3543 pixels, full page width: 7480 pixels).
- Combinations bitmapped line/halftones (color or grayscale): Save as TIFF, JPG or PNG files using a minimum of 500 dpi (for single column: min. 1772 pixels, full page width: 3740 pixels).

Please do not submit:

- Files that are too low in resolution.
- Artwork where text is disproportionately small compared to the image size, as text may become unreadable.
- Different images or graphs combined into one, as this affects accessibility

Color artwork

If you submit usable color figures with your accepted article, we will ensure that they appear in color online.

Please ensure that color images are accessible to all, including those with impaired color vision. Learn more about [color and web accessibility](#).

Generative AI and Figures, images and other artwork

Please read our policy on the use of generative AI and AI-assisted tools in figures, images and artwork, which can be found in Elsevier's [GenAI Policies for Journals](#). This policy states:

- AI tools may be used to support the creation of explanatory images (such as flow charts, conceptual diagrams, or schematic illustrations)
- AI tools may be used to create data visualizations (such as plots, charts, graphs, or heatmaps) when the output is directly derived from underlying data using reproducible analytical, computational, or statistical methods.
- AI tools must not be used to create or alter images that represent primary observed or experimental data (such as western blots, microscopy images, histology images, or patient images) that were not directly obtained in the research. This includes adjustments to brightness, contrast, or color balance, which should only be done using established image processing software.
- This policy does not prevent the use of AI tools in formal research design or research methods (such as in AI-assisted imaging approaches used for collection and/or interpretation of the underlying research data or predictive modelling of clinical outcomes based on patient data).

Authors are responsible for ensuring the accuracy and originality of all images submitted for publication. If you used generative AI tools to create images in your manuscript, you should disclose this in each image caption as well as in the general Generative AI disclosure statement.

Supplementary material

We encourage the use of supplementary materials such as applications, images and sound clips to enhance research. Some guidelines:

- Supplementary material should be accurate and relevant to the research.
- Cite all supplementary files in the manuscript text.
- Submit all supplementary materials at the same time as your manuscript.
- Include a concise, descriptive caption for each supplementary file, describing its content.
- After submission supplementary files can only be added or replaced in the revision stage of the editorial process.
- Be aware that all supplementary materials provided will appear online in the exact same way as received. These files will not be checked, formatted or typeset by the production team.

Video

This journal accepts video material and animation sequences to support and enhance your scientific research. We encourage you to include links to video or animation files within articles. Some guidelines:

- When including video or animation file links within your article, refer to the video or animation content by adding a note in your text where the file should be placed.
- Clearly label files ensuring the given file name is directly related to the file content.
- Provide files in one of our [recommended file formats](#). Files should be within our preferred maximum file size of 150 MB per file, 1 GB in total.
- Provide "stills" for each of your files. These will be used as standard icons to personalize the link to your video data. You can choose any frame from your video or animation or make a separate image.
- Provide descriptive text in your manuscript to refer to the video content. This text helps ensure accessibility for visually impaired readers who rely on descriptive information. For journals publishing in print this is also essential, as video and animation files cannot be embedded in the print version.

We publish all video and animation files supplied in the electronic version of your article.

For more detailed instructions, we recommend that you read our guidelines on [submitting video content to be included in the body of an article](#).

Research data

We are committed to supporting the storage of, access to and discovery of research data, and our [research data policy](#) sets out the principles guiding how we work with the research community to support a more efficient and transparent research process.

Research data refers to the results of observations or experimentation that validate research findings, which may also include software, code, models, algorithms, protocols, methods and other useful materials related to the project.

Please read our guidelines on [sharing research data](#) for more information on depositing, sharing and using research data and other relevant research materials.

Research data deposit, citation and linking

For this journal, **Option B** instructions from our [research data guidelines](#) apply. This means that you are **encouraged** to:

- Deposit your research data in a relevant data repository.
- Cite and link to this dataset in your article.
- If this is not possible, make a statement explaining why research data cannot be shared.

Data statement

To foster transparency, you are encouraged to state the availability of any data at submission.

Ensuring data is available may be a requirement of your funding body or institution. If your data is unavailable to access or unsuitable to post, you can state the reason why (e.g., your research data includes sensitive or confidential information such as patient data) during the submission process. This statement will appear with your published article on ScienceDirect.

Read more about the importance and benefits of providing a [data statement](#).

Data linking

Linking to the data underlying your work increases your exposure and may lead to new collaborations. It also provides readers with a better understanding of the described research.

If your research data has been made available in a data repository there are a number of ways your article can be linked directly to the dataset:

- Provide a link to your dataset when prompted during the online submission process.
- For some data repositories, a repository banner will automatically appear next to your published article on ScienceDirect.
- You can also link relevant data or entities within the text of your article through the use of identifiers. Use the following format: Database: 12345 (e.g. TAIR: AT1G01020; CCDC: 734053; PDB: 1XFN).

Learn more about [linking research data and research articles in ScienceDirect](#).

Co-submission of related data, methods or protocols

You are encouraged to publish a description of your research data, methods, or protocols related to your regular article as a co-submission article in [Data in Brief](#) or [MethodsX](#).

This co-submission can be submitted at the same time as your regular manuscript submission to this journal. If both your regular and co-submitted manuscripts are accepted for publication, they will be linked together on ScienceDirect, thereby promoting research reproducibility, interoperability, and open science.

- Prepare a separate manuscript describing your research data, methods, or protocols using the **mandatory templates** from [Data in Brief](#) or in [MethodsX](#).

- Submit this co-submission alongside your regular manuscript as a separate submission item; on the '**Attach files**' page in Editorial Manager, after which it will be automatically forwarded to the *Data in Brief* or *MethodsX* and assigned an editor.
- For associated Article Processing Charge (APC) details, please visit:
 - [MethodsX](#)
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Please note: Due to the automated process of submitting to *Data in Brief* or *MethodsX*, only after acceptance will you be presented with a personalized OA Article Publishing Charge based on your individual context (your country, institutional affiliation, and any society membership for example).

Please visit [co-submission Researcher Support page](#) for more information on how to submit a co-submission.

Journal specific information: Writing and formatting

Covering Letter

The letter must contain: why the submission is appropriate for publication in Computer Methods and Programs in Biomedicine; what is known about the topic discussed; what your study adds; and confirmation that the paper has not been published elsewhere.

Title Page

A separate sheet should include the title, the names and full addresses of the authors, and the name, full postal and email address and telephone number of the author for correspondence. Title should not be all capitals. *The Original research manuscript*

Manuscripts should be typed in English with double spacing and wide margins. Should not normally exceed 3500 words excluding abstract of 350 words. References up to 50 and no specific limits for Tables and Figures. Greek letters and mathematical symbols should be defined initially in the margin.

Abstract. This should be structured under the following headings: Background and Objectives, Methods, Results, Conclusions.

The Abstract of the manuscript should not exceed 350 words and must be structured into separate sections: **Background and Objective:** the context and purpose of the study; **Methods:** how the study was performed and statistical tests used; **Results:** the main findings including evaluation with precision, sensitivity and specificity values; **Conclusions:** brief summary and potential implications. Please minimize the use of abbreviations and do not cite references in the abstract.

Paper Sections: • Introduction • Methods • Results • Discussion • Acknowledgements including declarations: Statements of ethical approval, funding and competing interests. (Discussion should be separated from results with an updated literature).

Review articles

Systematic reviews of the literature on topics of broad interest to the readers or target audience and scope (covered topics). The structured abstract and text for a systematic review should follow the same format as described above with the word count: up to 4000 words excluding abstract of 350 words, Tables and Figures in total up to 5 with unlimited References.

Letters to the Editor

Letters to the Editor are considered for publication as provided they do not contain material that has been submitted or published elsewhere.

Letters in reference to a Journal article must not exceed 1000 words (excluding references), and must be received within three weeks after publication of the article.

Letters not related to a Journal article must not exceed 1500 words (excluding references) and contains no more than five references and one figure or table.

Article structure

Article sections

- Divide your manuscript into clearly defined and numbered sections. Number subsections 1.1 (then 1.1.1, 1.1.2, ...), then 1.2, etc.
- Use the numbering format when cross-referencing within your article. Do not just refer to "the text."
- You may give subsections a brief heading. Headings should appear on a separate line.
- Do not include the article abstract within section numbering.

Theory and calculation

The theory section should lay the foundation for further work by extending the background you provided in the introduction to your article. The calculation section should represent a practical development from a theoretical basis.

Acknowledgements

Include any individuals who provided you with help during your research, such as help with language, writing or proof reading, in the acknowledgements section.

Acknowledgements should be placed in a separate section which appears directly before the reference list. Do not include acknowledgements on your title page, as a footnote to your title, or anywhere else in your article other than in the separate acknowledgements section.

Author contributions: CRediT

Corresponding authors are required to acknowledge co-author contributions using [CRediT \(Contributor Roles Taxonomy\)](#) roles:

- Conceptualization
- Data curation

- Formal analysis
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- Software
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- Validation
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- Writing – original draft
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We advise you to read [more about CRediT](#) and [view an example of a CRediT author statement](#).

Appendices

We ask you to use the following format for appendices:

- Identify individual appendices within your article using the format: A, B, etc.
- Give separate numbering to formulae and equations within appendices using formats such as Eq. (A.1), Eq. (A.2), etc. and in subsequent appendices, Eq. (B.1), Eq. (B. 2) etc. In a similar way, give separate numbering to tables and figures using formats such as Table A.1; Fig. A.1, etc.

References

References within text

Any references cited within your article should also be present in your reference list and vice versa. Some guidelines:

- References cited in your abstract must be given in full.
- We recommend that you do not include unpublished results and personal communications in your reference list, though you may mention them in the text of your article.

- Any unpublished results and personal communications included in your reference list must follow the standard reference style of the journal. In substitution of the publication date add "unpublished results" or "personal communication."
- References cited as "in press" imply that the item has been accepted for publication.

Linking to cited sources will increase the discoverability of your research. We encourage the use of Digital Object Identifiers (DOIs) as reference links, as they provide a permanent link to the electronic article referenced.

Before submission, check that all data provided in your reference list are correct, including any references which have been copied. Providing correct reference data allows us to link to abstracting and indexing services such as Scopus, Crossref and PubMed. Any incorrect surnames, journal or book titles, publication years or pagination within your references may prevent link creation.

Reference format

This journal does not set strict requirements on reference formatting at submission. Some guidelines:

- References can be in any style or format as long as the style is consistent.
- Author names, journal or book titles, chapter or article titles, year of publication, volume numbers, article numbers or pagination must be included, where applicable.

Our journal reference style will be applied to your article after acceptance, at proof stage. If required, at this stage we will ask you to correct or supply any missing reference data.

Reference style

Indicate references by adding a number within square brackets in the text. You can refer to author names within your text, but you must always give the reference number, e.g., "as demonstrated [3,6]. Barnaby and Jones [8] obtained a different result".

Number references in the order they appear in your article.

Abbreviate journal names according to the [List of Title Word Abbreviations](#) (LTWA).

Examples:

Reference to a journal publication:

[1] J. van der Geer, T. Handgraaf, R.A. Lupton, The art of writing a scientific article, J. Sci. Commun. 163 (2020) 51 – 59.
<https://doi.org/10.1016/j.sc.2020.00372>.

Reference to a journal publication with an article number:

[2] J. van der Geer, T. Handgraaf, R.A. Lupton, 2022. The art of writing a scientific article. Heliyon. 19, e00205.
<https://doi.org/10.1016/j.heliyon.2022.e00205>.

Reference to a book:

[3] W. Strunk Jr., E.B. White, The Elements of Style, fourth ed., Longman, New York, 2000.

Reference to a chapter in a book:

[4] G.R. Mettam, L.B. Adams, How to prepare an electronic version of your article, in: B.S. Jones, R.Z. Smith (Eds.), Introduction to the Electronic Age, E-Publishing Inc., New York, 2020, pp. 281 - 304.

Reference to a website:

[5] Cancer Research UK, Cancer statistics reports for the UK.

<http://www.cancerresearchuk.org/aboutcancer/statistics/cancerstatsreport/>, 2023 (accessed 13 March 2023).

Reference to a dataset:

[6] M. Oguro, S. Imahiro, S. Saito, T. Nakashizuka, Mortality data for Japanese oak wilt disease and surrounding forest compositions [dataset], Mendeley Data, v1, 2015. <https://doi.org/10.1234/abc12nb39r.1>.

Reference to software:

[7] E. Coon, M. Berndt, A. Jan, D. Svyatsky, A. Atchley, E. Kikinson, D. Harp, G. Manzini, E. Shelef, K. Lipnikov, R. Garimella, C. Xu, D. Moulton, S. Karra, S. Painter, E. Jafarov, S. Molins, Advanced Terrestrial Simulator (ATS) v0.88 [software], Zenodo, March 25, 2020. <https://doi.org/10.1234/zenodo.3727209>.

Web references

When listing web references, as a minimum you should provide the full URL and the date when the reference was last accessed. Additional information (e.g. DOI, author names, dates or reference to a source publication) should also be provided, if known.

You can list web references separately under a new heading directly after your reference list or include them in your reference list.

Data references

We encourage you to cite underlying or relevant datasets within article text and to list data references in the reference list.

When citing data references, you should include:

- author name(s)
- dataset title
- data repository
- version (where available)
- year
- global persistent identifier

Add [dataset] immediately before your reference. This will help us to properly identify the dataset. The [dataset] identifier will not appear in your published article.

Preprint references

We ask you to mark preprints clearly. You should include the word "preprint" or the name of the preprint server as part of your reference and provide the preprint DOI.

Where a preprint has subsequently become available as a peer-reviewed publication, use the formal publication as your reference.

If there are preprints that are central to your work or that cover crucial developments in the topic, but they are not yet formally published, you may reference the preprint.

Reference management software

Most journals have their reference template available in popular reference management software products. These include products that support [Citation Style Language \(CSL\)](#) such as [Mendeley Reference Manager](#).

If you use a citation plug-in from these products, select the relevant journal template and all your citations and bibliographies will automatically be formatted in the journal style. We advise you to [remove all field codes](#) before submitting your manuscript to any reference management software product.

If a template is not available for this journal, follow the format given in examples in the reference style section of this Guide for Authors.

Submitting your manuscript

Submission checklist

Before completing the submission of your manuscript, we advise you to read our submission checklist:

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- All files have been uploaded, including artwork, videos, and supplementary materials. Don't forget to include tables and footnotes in the files, and to submit captions.
- Spelling and grammar checks have been carried out.
- All references in the article text are cited in the reference list, and vice versa.
- Permission has been obtained for the use of any copyrighted material from other sources, including the Web.
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El objetivo de la revisión es comprobar la maquetación, la edición, la integridad y la corrección del texto, las tablas y las figuras de su artículo. Los cambios significativos realizados durante la fase de revisión solo se considerarán con la aprobación del editor de la revista.

Compartir enlace

Se enviará por correo electrónico al autor correspondiente un enlace personalizado para compartir el artículo, que le dará acceso gratuito durante 50 días a la versión final publicada en [ScienceDirect](#). Este enlace podrá utilizarse para compartir el artículo en cualquier canal de comunicación, como correo electrónico o redes sociales.

No se proporcionará un enlace para compartir si su artículo se publica en acceso abierto. La versión final publicada de su artículo de acceso abierto estará disponible públicamente en ScienceDirect y podrá compartirse a través del enlace DOI del artículo.

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Academia de Investigadores de Elsevier

Si desea ayuda para mejorar su envío o para desenvolverse en el proceso de publicación, puede obtener asistencia a través de [Elsevier Researcher Academy](#).

Elsevier Researcher Academy ofrece módulos de aprendizaje electrónico gratuitos, seminarios web, guías descargables y recursos sobre redacción de investigaciones y procesos de revisión por pares.

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Le recomendamos que escriba en inglés americano o británico, pero no una combinación de ambos.

Si considera que el idioma inglés de su manuscrito necesita ser revisado para eliminar posibles errores gramaticales u ortográficos y para ajustarse al inglés científico correcto, puede utilizar el [servicio de edición de idioma inglés](#) que ofrece Elsevier Author Services.

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